9.14 Stokes' Theorem

Introduction Green's theorem of the preceding section has two vector forms. In this and in Section 9.16 we shall generalize these forms to three dimensions.

Vector Form of Green's Theorem If $\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ is a two-dimensional vector field, then

$$\text{curl}\,\mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & 0 \end{vmatrix} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}.$$

From (12) and (13) of Section 9.8, Green's theorem can be written in vector notation as

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C \mathbf{F} \cdot \mathbf{T}\, ds = \iint_R (\text{curl}\,\mathbf{F}) \cdot \mathbf{k}\, dA; \tag{1}$$

that is, the line integral of the tangential component of $\mathbf{F}$ is the double integral of the normal component of curl $\mathbf{F}$.

Green's Theorem in 3-Space The vector form of Green's theorem given in (1) relates a line integral around a piecewise-smooth simple closed curve C forming the boundary of a plane region R to a double integral over R. Green's theorem in 3-space relates a line integral around a piecewise-smooth simple closed curve C forming the boundary of a surface S with a surface integral over S. Suppose $z = f(x, y)$ is a continuous function whose graph is a piecewise-smooth orientable surface over a region R on the xy-plane. Let C form the boundary of S and let the projection of C onto the xy-plane form the boundary of R. The positive direction on C is induced by the orientation of the surface S; the positive direction on C corresponds to the direction a person would have to walk on C to have his or her head point in the direction of the orientation of the surface while keeping the surface to the left. See FIGURE 9.14.1. More precisely, the positive orientation of C is in accordance with the right-hand rule: If the thumb of the right hand points in the direction of the orientation of the surface, then roughly the fingers of the right hand wrap around the surface in the positive direction. Finally, let $\mathbf{T}$ be a unit tangent vector to C that points in the positive direction. The three-dimensional form of Green's theorem, which we now give, is called **Stokes' theorem** after the Irish mathematical physicist **George G. Stokes** (1819–1903).

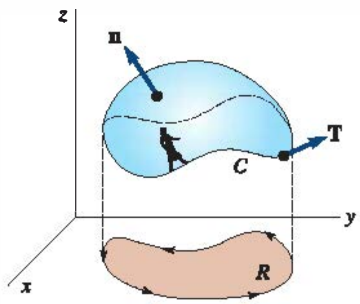

FIGURE 9.14.1 Boundary C of surface S has positive orientation

Theorem 9.14.1 Stokes' Theorem

Let S be a piecewise-smooth orientable surface bounded by a piecewise-smooth simple closed curve C. Let $\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$ be a vector field for which P, Q, and R are continuous and have continuous first partial derivatives in a region of 3-space containing S. If C is traversed in the positive direction, then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C (\mathbf{F} \cdot \mathbf{T})\, dS = \iint_S (\text{curl}\,\mathbf{F}) \cdot \mathbf{n}\, dS, \tag{2}$$

where $\mathbf{n}$ is a unit normal to S in the direction of the orientation of S.

PARTIAL PROOF: Suppose the surface S is oriented upward and is defined by a function $z = f(x, y)$ that has continuous second partial derivatives. From Definition 9.7.1 we have

$$\text{curl}\,\mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}.$$

Furthermore, if we write $g(x, y, z) = z - f(x, y) = 0$, then

$$\mathbf{n} = \frac{\nabla g}{\|\nabla g\|} = \frac{-\dfrac{\partial f}{\partial x}\mathbf{i} - \dfrac{\partial f}{\partial y}\mathbf{j} + \mathbf{k}}{\sqrt{1 + \left(\dfrac{\partial f}{\partial x}\right)^2 + \left(\dfrac{\partial f}{\partial y}\right)^2}}.$$

Hence,

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n}\, dS = \iint_R \left[-\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\frac{\partial f}{\partial x} - \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\frac{\partial f}{\partial y} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\right] dA. \quad (3)$$

Our goal is now to show that $\oint_C \mathbf{F} \cdot d\mathbf{r}$ reduces to (3).

If C_{xy} is the projection of C onto the xy-plane and has the parametric equations $x = x(t)$, $y = y(t)$, $a \le t \le b$, then parametric equations for C are $x = x(t)$, $y = y(t)$, $z = f(x(t), y(t))$, $a \le t \le b$. Thus,

$$\begin{aligned}
\oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C P\,dx + Q\,dy + R\,dz \\
&= \int_a^b \left[P\frac{dx}{dt} + Q\frac{dy}{dt} + R\left(\frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}\right)\right] dt \qquad \leftarrow \text{Chain Rule} \\
&= \oint_{C_{xy}} \left(P + R\frac{\partial f}{\partial x}\right) dx + \left(Q + R\frac{\partial f}{\partial y}\right) dy \qquad (4) \\
&= \iint_R \left[\frac{\partial}{\partial x}\left(Q + R\frac{\partial f}{\partial y}\right) - \frac{\partial}{\partial y}\left(P + R\frac{\partial f}{\partial x}\right)\right] dA. \qquad \leftarrow \text{Green's theorem}
\end{aligned}$$

Now,

$$\begin{aligned}
\frac{\partial}{\partial x}\left(Q + R\frac{\partial f}{\partial y}\right) &= \frac{\partial}{\partial x}\left[Q(x, y, f(x, y)) + R(x, y, f(x, y))\frac{\partial f}{\partial y}\right] \\
&= \frac{\partial Q}{\partial x} + \frac{\partial Q}{\partial z}\frac{\partial f}{\partial x} + R\frac{\partial^2 f}{\partial x\,\partial y} + \frac{\partial f}{\partial y}\left(\frac{\partial R}{\partial x} + \frac{\partial R}{\partial z}\frac{\partial f}{\partial x}\right) \qquad \leftarrow \text{Chain and Product Rules} \quad (5) \\
&= \frac{\partial Q}{\partial x} + \frac{\partial Q}{\partial z}\frac{\partial f}{\partial x} + R\frac{\partial^2 f}{\partial x\,\partial y} + \frac{\partial R}{\partial x}\frac{\partial f}{\partial y} + \frac{\partial R}{\partial z}\frac{\partial f}{\partial y}\frac{\partial f}{\partial x}.
\end{aligned}$$

Similarly,

$$\frac{\partial}{\partial y}\left(P + R\frac{\partial f}{\partial x}\right) = \frac{\partial P}{\partial y} + \frac{\partial P}{\partial z}\frac{\partial f}{\partial y} + R\frac{\partial^2 f}{\partial y\,\partial x} + \frac{\partial R}{\partial y}\frac{\partial f}{\partial x} + \frac{\partial R}{\partial z}\frac{\partial f}{\partial x}\frac{\partial f}{\partial y}. \quad (6)$$

Subtracting (6) from (5) and using the fact that $\partial^2 f/\partial x\partial y = \partial^2 f/\partial y\partial x$, we see that (4) becomes, after rearranging,

$$\iint_R \left[-\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\frac{\partial f}{\partial x} - \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\frac{\partial f}{\partial y} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\right] dA.$$

This last expression is the same as the right side of (3), which was to be shown. ≡

EXAMPLE 1 Verifying Stokes' Theorem

Let S be the part of the cylinder $z = 1 - x^2$ for $0 \le x \le 1$, $-2 \le y \le 2$. Verify Stokes' theorem for the vector field $\mathbf{F} = xy\mathbf{i} + yz\mathbf{j} + xz\mathbf{k}$. Assume S is oriented upward.

SOLUTION The surface S, the curve C (which is composed of the union of C_1, C_2, C_3, and C_4), and the region R are shown in **FIGURE 9.14.2**.

Surface Integral: From $\mathbf{F} = xy\mathbf{i} + yz\mathbf{j} + xz\mathbf{k}$, we find

$$\operatorname{curl} \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ xy & yz & xz \end{vmatrix} = -y\mathbf{i} - z\mathbf{j} - x\mathbf{k}.$$

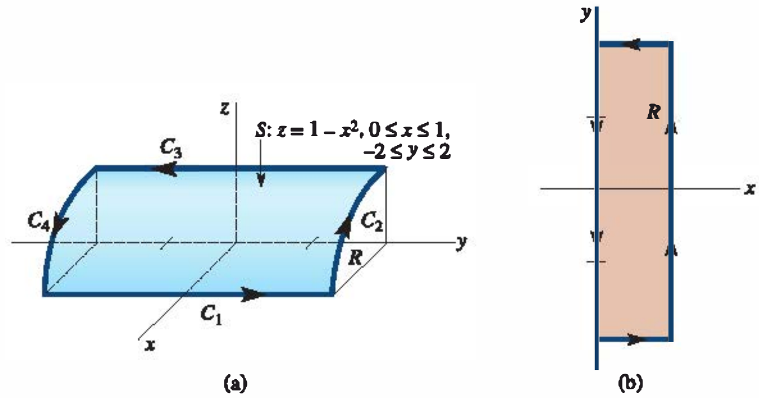

FIGURE 9.14.2 Surface S and region R in Example 1

Now, if $g(x, y, z) = z + x^2 - 1 = 0$ defines the cylinder, then the upper normal is

$$\mathbf{n} = \frac{\nabla g}{\|\nabla g\|} = \frac{2x\mathbf{i} + \mathbf{k}}{\sqrt{4x^2 + 1}}.$$

Therefore, $$\iint_S (\operatorname{curl} \mathbf{F} \cdot \mathbf{n})\, dS = \iint_S \frac{-2xy - x}{\sqrt{4x^2 + 1}}\, dS.$$

To evaluate the latter surface integral, we use (5) of Section 9.13:

$$\begin{aligned} \iint_S \frac{-2xy - x}{\sqrt{4x^2 + 1}}\, dS &= \iint_R (-2xy - x)\, dA \\ &= \int_0^1 \int_{-2}^{2} (-2xy - x)\, dy\, dx \\ &= \int_0^1 \Big[-xy^2 - xy \Big]_{-2}^{2} dx \\ &= \int_0^1 (-4x)\, dx = -2. \end{aligned} \qquad (7)$$

Line Integral: We write $\oint_C = \int_{C_1} + \int_{C_2} + \int_{C_3} + \int_{C_4}$.

On C_1: $x = 1, z = 0, dx = 0, dz = 0$, so

$$\int_{C_1} y(0) + y(0)\, dy + 0 = 0.$$

On C_2: $y = 2, z = 1 - x^2, dy = 0, dz = -2x\, dx$, so

$$\int_{C_2} 2x\, dx + 2(1 - x^2)0 + x(1 - x^2)(-2x\, dx) = \int_1^0 (2x - 2x^2 + 2x^4)\, dx = -\frac{11}{15}.$$

On C_3: $x = 0, z = 1, dx = 0, dz = 0$, so

$$\int_{C_3} 0 + y\,dy + 0 = \int_{2}^{-2} y\,dy = 0.$$

On C_4: $y = -2, z = 1 - x^2, dy = 0, dz = -2x\,dx$, so

$$\int_{C_4} -2x\,dx - 2(1 - x^2)0 + x(1 - x^2)(-2x\,dx) = \int_0^1 (-2x - 2x^2 + 2x^4)\,dx = -\frac{19}{15}.$$

Hence, $$\oint_C xy\,dx + yz\,dy + xz\,dz = 0 - \frac{11}{15} + 0 - \frac{19}{15} = -2,$$

which, of course, agrees with (7). ≡

EXAMPLE 2 Using Stokes' Theorem

Evaluate $\oint_C z\,dx + x\,dy + y\,dz$, where C is the trace of the cylinder $x^2 + y^2 = 1$ in the plane $y + z = 2$. Orient C counterclockwise as viewed from above. See FIGURE 9.14.3.

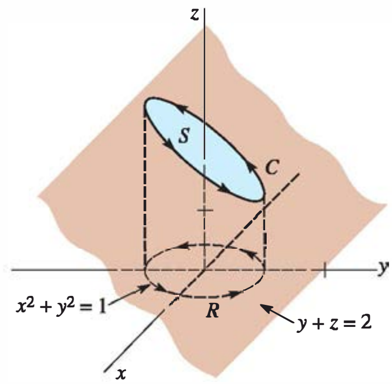

FIGURE 9.14.3 Curve C in Example 2

SOLUTION If $\mathbf{F} = z\,\mathbf{i} + x\mathbf{j} + y\mathbf{k}$, then

$$\text{curl }\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ z & x & y \end{vmatrix} = \mathbf{i} + \mathbf{j} + \mathbf{k}.$$

The given orientation of C corresponds to an upward orientation of the surface S. Thus, if $g(x, y, z) = y + z - 2 = 0$ defines the plane, then the upper normal is

$$\mathbf{n} = \frac{\nabla g}{\|\nabla g\|} = \frac{1}{\sqrt{2}}\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}.$$

Hence, from (2),

$$\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_S \left[(\mathbf{i} + \mathbf{j} + \mathbf{k})\cdot\left(\frac{1}{\sqrt{2}}\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}\right)\right] dS$$

$$= \sqrt{2}\iint_S dS = \sqrt{2}\iint_R \sqrt{2}\,dA = 2\pi.$$ ≡

Note that if $\mathbf{F}$ is the gradient of a scalar function, then, in view of (5) in Section 9.7, (2) implies that the circulation $\oint_C \mathbf{F}\cdot d\mathbf{r}$ is zero. Conversely, it can be shown that if the circulation is zero for every simple closed curve, then $\mathbf{F}$ is the gradient of a scalar function. In other words, $\mathbf{F}$ is irrotational if and only if $\mathbf{F} = \nabla\phi$, where ϕ is a potential for $\mathbf{F}$. Equivalently, this gives a test for a conservative vector field:

$\mathbf{F}$ is a conservative vector field if and only if curl $\mathbf{F} = \mathbf{0}$.

Physical Interpretation of Curl In Section 9.8 we saw that if $\mathbf{F}$ is a velocity field of a fluid, then the circulation $\oint_C \mathbf{F}\cdot d\mathbf{r}$ of $\mathbf{F}$ around C is a measure of the amount by which the fluid tends to turn the curve C by circulating around it. The circulation of $\mathbf{F}$ is closely related to the curl of $\mathbf{F}$. To see this, suppose $P_0(x_0, y_0, z_0)$ is any point in the fluid and C_r is a small circle of radius r centered at P_0. See FIGURE 9.14.4. Then by Stokes' theorem,

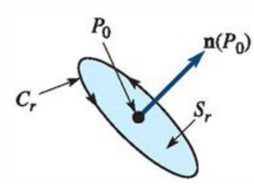

FIGURE 9.14.4 Curve C_r and surface S_r in (8)

$$\oint_{C_r} \mathbf{F}\cdot d\mathbf{r} = \iint_{S_r} (\text{curl }\mathbf{F})\cdot\mathbf{n}\,dS. \tag{8}$$

Now at all points $P(x, y, z)$ within the small circle C_r, if we take $\operatorname{curl}\mathbf{F}(P) \approx \operatorname{curl}\mathbf{F}(P_0)$, then (8) gives the approximation

$$\begin{aligned} \oint_{C_r} \mathbf{F} \cdot d\mathbf{r} &\approx \iint_{S_r} (\operatorname{curl}\mathbf{F}(P_0)) \cdot \mathbf{n}(P_0)\, dS \\ &= (\operatorname{curl}\mathbf{F}(P_0)) \cdot \mathbf{n}(P_0) \iint_{S_r} dS \qquad (9) \\ &= (\operatorname{curl}\mathbf{F}(P_0)) \cdot \mathbf{n}(P_0)\, A_r, \end{aligned}$$

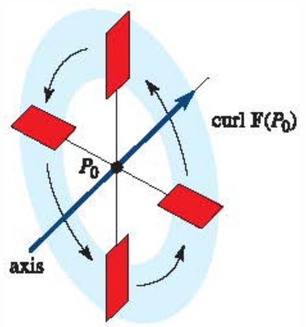

FIGURE 9.14.5 Paddle wheel

where A_r is the area (πr^2) of the circular surface S_r. As we let $r \to 0$, the approximation $\operatorname{curl}\mathbf{F}(P) \approx \operatorname{curl}\mathbf{F}(P_0)$ becomes better, and so (9) yields

$$(\operatorname{curl}\mathbf{F}(P_0)) \cdot \mathbf{n}(P_0) = \lim_{r \to 0} \frac{1}{A_r} \oint_{C_r} \mathbf{F} \cdot d\mathbf{r}. \qquad (10)$$

Thus we see that the normal component of curl **F** is the limiting value of the ratio of the circulation of **F** to the area of the circular surface. For a small but fixed value of r, we have

$$(\operatorname{curl}\mathbf{F}(P_0)) \cdot \mathbf{n}(P_0) \approx \frac{1}{A_r} \oint_{C_r} \mathbf{F} \cdot d\mathbf{r}. \qquad (11)$$

Roughly then, the curl of **F** is the circulation of **F** per unit area. If $\operatorname{curl}\mathbf{F}(P_0) \neq \mathbf{0}$, then the left-hand side of (11) is a maximum when the circle C_r is situated in a manner so that $\mathbf{n}(P_0)$ points in the same direction as $\operatorname{curl}\mathbf{F}(P_0)$. In this case, the circulation on the right side of (11) will also be a maximum. Thus, a paddle wheel inserted into the fluid at P_0 will rotate fastest when its axis points in the direction of $\operatorname{curl}\mathbf{F}(P_0)$. See **FIGURE 9.14.5**. Note, too, that the paddle wheel will not rotate if its axis is perpendicular to $\operatorname{curl}\mathbf{F}(P_0)$.

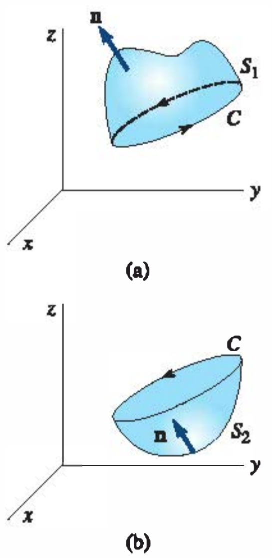

FIGURE 9.14.6 Two surfaces with same boundary C

Remarks

The value of the surface integral in (2) is determined solely by the integral around its boundary C. This basically means that the shape of the surface S is irrelevant. Assuming that the hypotheses of Theorem 9.14.1 are satisfied, then for two different surfaces S_1 and S_2 with the same orientation and with the same boundary C, we have

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_{S_1} (\operatorname{curl}\mathbf{F}) \cdot \mathbf{n}\, dS = \iint_{S_2} (\operatorname{curl}\mathbf{F}) \cdot \mathbf{n}\, dS.$$

See **FIGURE 9.14.6** and Problems 17 and 18 in Exercises 9.14.

9.14 Exercises

Answers to selected odd-numbered problems begin on page ANS-24.

In Problems 1–4, verify Stokes' theorem. Assume that the surface S is oriented upward.

1. $\mathbf{F} = 5y\mathbf{i} - 5x\mathbf{j} + 3\mathbf{k}$; S that portion of the plane $z = 1$ within the cylinder $x^2 + y^2 = 4$
2. $\mathbf{F} = 2z\mathbf{i} - 3x\mathbf{j} + 4y\mathbf{k}$; S that portion of the paraboloid $z = 16 - x^2 - y^2$ for $z \geq 0$
3. $\mathbf{F} = z\mathbf{i} + x\mathbf{j} + y\mathbf{k}$; S that portion of the plane $2x + y + 2z = 6$ in the first octant
4. $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$; S that portion of the sphere $x^2 + y^2 + z^2 = 1$ for $z \geq 0$

In Problems 5–12, use Stokes' theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$. Assume C is oriented counterclockwise as viewed from above.

5. $\mathbf{F} = (2z + x)\mathbf{i} + (y - z)\mathbf{j} + (x + y)\mathbf{k}$; C the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$
6. $\mathbf{F} = z^2y\cos xy\,\mathbf{i} + z^2x(1 + \cos xy)\mathbf{j} + 2z\sin xy\,\mathbf{k}$; C the boundary of the plane $z = 1 - y$ shown in **FIGURE 9.14.7**.